NAME

mbpy_grdtiff − Create an executable shellscript which will generate a georeferenced TIFF image of gridded data using GMT modern mode.

VERSION

Version 5.0

SYNOPSIS

mbpy_grdtiff **−I***grid* [**−O***root*] [**−X**] [**−V**]
[**−R***west/east/south/north*] [**−N***dx/dy*]
[**−G***color_mode*] [**−W***color_style/palette*[*/ncolors*] | *cptfile*] [**−D**] [**−Y**] [**−S**] [**−Z***zmin/zmax*[*/mode*]]
[**−−data−scale** *factor*]
[**−A***magnitude*[*/azimuth*[*/elevation*]]] [**−−magnitude** *val*] [**−−azimuth** *deg*] [**−−elevation** *deg*] [**−−shade−flip**]
[**−K***intensity_grid*] [**−−stretch−shade**]
[**−−gmt−default** *param/value*]

DESCRIPTION

mbpy_grdtiff is a Python/PyGMT-era replacement for the Perl macro **mbm_grdtiff**. It reads a grid (for instance one made by **mbgrid** or **mbmosaic**) and generates an executable shellscript of **GMT modern mode** commands which, when run, makes a georeferenced TIFF image of the grid with the **MB-System** GMT module **mbgrdtiff**. The image is a GeoTIFF with the grid's coordinate system embedded, plus a parallel ".tfw" world file for software that does not read the GeoTIFF tags. Geographic grids and grids in projected coordinate systems (e.g. UTM) are both handled. Such images can be loaded into GIS packages such as QGIS and ArcGIS as geographically located coverages.

The image may be a color fill of the gridded values, color shaded relief, a color fill shaded by a separate intensity grid, a color fill of the slope magnitude, or a color fill shaded by the slope magnitude (see **−G**). The color palettes, color intervals and shading methods are the same as those of **mbpy_grdplot**, so a GeoTIFF and a map of the same grid made with the same options look the same.

The generated script is named *root*_tiff.sh and creates *root*.tif and *root*.tfw. A generated color palette is written to *root*.cpt by **mbpy_grdtiff** itself. Because **mbgrdtiff** writes its image file directly, the script opens a GMT modern-mode session without a figure (**gmt begin** ... **gmt end**) only so that its temporary GMT defaults stay inside the session; no gmt.conf file is created or deleted in the current directory.

Differences from **mbm_grdtiff**: a file listing several grids is not accepted by **−I** (use one **mbpy_grdtiff** command per grid); the **−MGW** option is not needed because **mbgrdtiff** always writes the world file; the **−MGS** option is **−−data−scale**; the sealevel palettes 8 and 9 of **mbpy_grdplot** are available; discrete palettes and the histogram-equalized color stretch follow **mbm_grdplot** and **mbpy_grdplot**; and the image is not displayed after it is made.

mbpy_grdtiff requires **GMT** 6 or later with the **MB-System** GMT modules, and a Python 3 interpreter. It uses the modules **mbpy_layout.py**, **mbpy_color.py**, **mbpy_misc.py**, **mbpy_grdplot_script.py** and **mbpy_grdtiff_script.py**, installed alongside it.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−I, **−−input** *grid*
Input grid file. Required.

−O, **−−output** *root*
Output root name: the script is *root*_tiff.sh, the image *root*.tif and the world file *root*.tfw. Default: the input grid's basename without its extension.

−X, **−−execute**
Run the generated script immediately after writing it.

−V, **−−verbose**
Print the generated script's path and a summary of the image style, region, z range and file names.

−R, **−−region** *west/east/south/north*
Bounds of the image, in the grid's own coordinates. When given, the color stretch spans the z range of the grid inside these bounds. Default: the grid's own bounds.

−N, **−−nudge** *dx/dy*
Shift the location of the image by *dx* meters east and *dy* meters north (**mbgrdtiff −N**).

−G, **−−color−mode** *color_mode*
Image style:
color_mode = 1: Color fill.
color_mode = 2: Color shaded relief, using an illumination model built from **−−magnitude**, **−−azimuth** and **−−elevation**.
color_mode = 3: Color fill shaded by the intensity grid given with **−K**.
color_mode = 4: Color fill of the slope magnitude.
color_mode = 5: Color fill shaded by the slope magnitude.
Default: 1, or 3 when **−K** is given.

−W, **−−color−control** *color_style/palette*[*/ncolors*] | *cptfile*
Color palette. *color_style* 1 gives a continuous palette, 2 a discrete one. *palette* is one of:
1: Haxby colors [default]
2: high intensity colors
3: low intensity colors
4: grayscale
5: uniform gray
6: uniform black
7: uniform white
8, 9: sealevel palettes, with a color break at zero (see **mbpy_grdplot**)
ncolors is the number of colors [11]. If the argument names an existing file, that file is used as the color palette table.

−D, **−−color−flip**
Reverse the color palette.

−Y, **−−no−nice−interval**
Stretch the colors linearly between the minimum and maximum values instead of using rounded color intervals.

−S, **−−stretch−color**
Use histogram equalization of the grid (**gmt grdhisteq**) to set the color boundaries. With palette 8 or 9 the values below and above zero are equalized separately.

−Z, **−−zbounds** *zmin/zmax*[*/mode*]
Use *zmin* and *zmax* instead of the grid's own z range for the color stretch. If *mode* is 1, the first and last colors are extended to the grid's own minimum and maximum, so that every value is colored.

−−data−scale *factor*
Multiply the grid values by *factor* before coloring and shading, for instance −1 to plot depths as topography (**mbm_grdtiff −MGS**).

−A, **−−shade** *magnitude*[*/azimuth*[*/elevation*]]
Shading parameters in one argument, as in **mbm_grdtiff −A**. The separate options below take precedence.

−−magnitude *val*
Shading magnitude. Defaults: 1.0 for modes 2, 4 and 5; −0.4 for mode 3.

−−azimuth *deg*
Illumination azimuth for mode 2, degrees clockwise from north [0].

−−elevation *deg*
Illumination elevation for mode 2, degrees above the horizon [30].

−−shade−flip
Reverse the sense of the shading (negates the magnitude).

−K, **−−intensity−file** *intensity_grid*
Intensity grid used to shade the image in mode 3; it must cover the same area as the input grid.

−−stretch−shade
Histogram-equalize the intensity grid before shading (mode 3).

−−gmt−default *param/value*
Override a GMT default parameter in the generated script; repeatable.

EXAMPLES

A color shaded relief GeoTIFF of a bathymetry grid, illuminated from the northeast, made at once:

```
mbpy_grdtiff −I ZTopo.grd −G2 −−azimuth 45 −X −V
```

A grayscale amplitude mosaic shaded by a separate intensity grid:

```
mbpy_grdtiff −I ZAmp.grd −O ZAmp_shaded −K ZTopoInt.grd \
   −W 1/4 −−stretch−shade
```

A topography image of a depth grid (positive down), with the colors clipped to 1000 to 4000 m depth but every value still colored:

```
mbpy_grdtiff −I depths.grd −−data−scale −1 −Z −4000/−1000/1
```

SEE ALSO

mbm_grdtiff(1), **mbgrdtiff**(1), **mbpy_grdplot**(1), **mbpy_plot**(1), **mbgrid**(1), **mbmosaic**(1), **mbsystem**(1)

BUGS

Please report any bugs to the MB-System project on GitHub (https://github.com/dwcaress/MB-System).